INCH-POUND

FF-B-2844/06
15 FEBRUARY 1994

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR: SINGLE ROW, RADIAL, NON-FILLING SLOT, EXTRA LIGHT SERIES (XLS - SERIES) (CHROMIUM ALLOY STEEL)

The General Services Administration has authorized the use of this specification by all Federal Agencies.

The complete requirements for procuring the bearings described herein shall consist of this document and the latest issue of FF-B-2844/GEN.

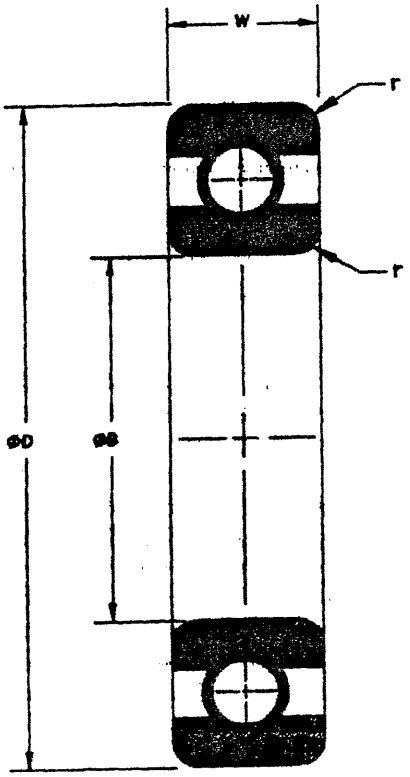


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)

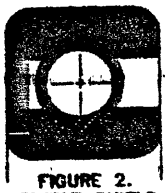


FIGURE 2.
SINGLE SHIELD



FIGURE 3.
DOUBLE SHIELD

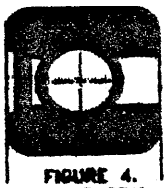


FIGURE 4.
SINGLE SEAL

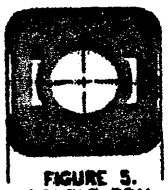


FIGURE 5.
DOUBLE SEAL



FIGURE 6.
SINGLE SHIELD AND SEAL

BEARING CONFIGURATIONS

FSC 3110

AMSC/NA

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REQUIREMENTS

1. **Boundary dimensions** The boundary dimensions shall be specified in the part number by the bore codes in Table I.

2. MATERIAL

2.1 Rings and balls. The rings and balls of bearings covered by this specification shall be made of chromium-alloy steel 52100 (UNS 52986).

2.2 Closures. The shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or non-removable type. Seals shall be impervious contact type. The number of shields and seals required shall be specified in the part number by using the code's listed in Table IV.

2.3 Cage material. Cages shall be manufactured from any material unless otherwise specified. The type of cage required shall be specified in the part number by using the codes listed in Table II.

3. METROLOGY

3.1 Radial internal clearance. The tolerance limits for radial internal clearance are specified-in ANSI/AFBMA STD 20 and shall be specified in the part number by using the codes listed in Table III.

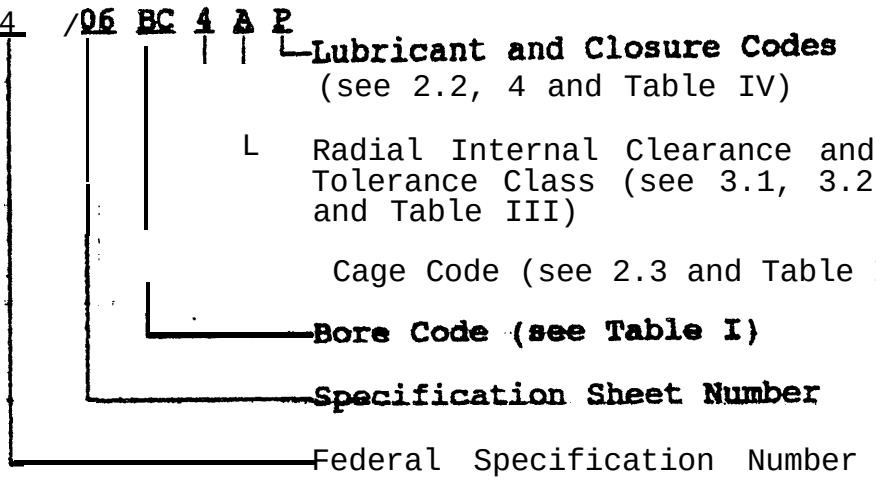
3.2 Tolerance class. The tolerance limits for tolerance class ARK-l, and ABEC 5, shall be in accordance with ANSI/AFEMA STD 20 and shall be specified in the part number by using the codes listed in Table 111.

4. Lubricant or preservative compound. The lubricant or preservative compound shall be specified in the part number by using the code listed in Table IV. Preservative compound shall be applied to all open beating (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings, the bearing part number shall include the "C", "F" or "L" suffix from Table IV. For double closure or single closure bearings that require lubricant, the bearing part number shall include the preferred suffix from Table IV as required. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled.

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5. Part number. The part number consists of the Federal Specification prefix letters FFB, the basic number of the Federal Specification, the specification sheet number, bore code, and alpha or numeric codes for cage, radial internal clearance and tolerance class, lubricant and closures.

EXAMPLE: FFB284 /06 BC 4 A P



FFB2844/06BC4AP indicates - Bearing Bore 1.6250 inches; outside diameter 2.8750 inches; width 0.5625 inches; cage type, machined non metallic; radial internal clearance, tight; tolerance-class, ABEC 1; single shield and single seal; lubricant in accordance with DoD-G-24508.

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TABLE I. BOUNDARY DIMENSIONS 1/

Bore Code	0B Bore Diameter (inch)	0B Bore Diameter (MM)	0D Outside Diameter	W Width	Chamfer Min. 2/	Dia. Series 3/	Dynamic Load Rating (lbs)
BA	1.3750	34.9250	2.5625	0.5625	.060	0	2786
BB	1.5000	38.1000	2.6875	0.5625	.060	0	2926
BC	1.6250	41.2750	2.8750	0.5625	.060	0	3050
BD	1.7500	44.4500	3.0000	0.5625	.060	0	3919
BE	1.8750	47.6250	3.1875	0.6250	.060	0	3878
CA	2.0000	50.8000	3.3125	0.6250	.060	0	4608
CB	2.1250	53.9750	3.4375	0.6250	.060	0	3995
CD	2.2500	57.1500	3.5625	0.6250	.060	0	4140
CE	2.3750	60.3250	3.7500	0.6875	.060	0	4950
CF	2.5000	63.5000	3.8750	0.6875	.060	0	4914
CG	2.6250	66.6750	4.1250	0.6875	.060	0	5045
CH	2.7500	69.8500	4.1250	0.6875	.060	0	5045
CJ	2.8750	73.0250	4.5000	0.7500	.090	0	6122
DA	3.0000	76.2000	4.5000	0.7500	.090	0	6122
DB	3.1250	79.3750	4.7500	0.7500	.090	0	6042
DC	3.2500	82.5500	4.7500	0.7500	.090	0	6042
DE	3.3750	85.7250	5.0000	0.7500	.090	0	7008
DF	3.5000	88.9000	5.0000	0.7500	.090	9	7008
DG	3.6250	92.0750	5.2500	0.7500	.090	0	7420
DH	3.7500	95.2500	5.2500	0.7500	.090	9	7420
DJ	3.8750	98.4250	5.6250	0.8750	.090	9	9087
EA	4.0000	101.6000	5.6250	0.8750	.090	9	8540
EB	4.1250	104.7750	6.0000	0.8750	.090	9	8363
EC	4.2500	107.9500	6.0000	0.8750	.090	9	8363
ED	4.3750	111.1250	6.2500	0.8750	.090	9	8270
EE	4.5000	114.3000	6.2500	0.8750	.090	9	8270
EF	4.6250	117.4750	6.5000	0.8750	.090	9	8730
EG	4.7500	120.6500	6.5000	0.8750	.090	9	8730
EH	4.8750	123.8250	7.0000	1.0000	.090	9	10,575
FA	5.0000	127.0000	7.0000	1.0000	.090	9	10,575
FB	5.1250	130.1750	7.2500	1.0000	.090	9	10,487
FC	5.2500	133.3500	7.2500	1.0000	.090	9	10,487
FD	5.3750	136.5250	7.5000	1.0000	.090	9	11,068
FE	5.5000	139.7000	7.5000	1.0000	.090	9	11,068
FF	5.6250	142.8750	7.7500	1.0000	.090	9	11,613
FG	5.7500	146.0500	7.7500	1.0000	.090	9	10,887
FH	5.8750	149.2250	8.0000	1.0000	.090	9	11,461
GA	6.0000	152.4000	8.0000	1.0000	.090	9	11,106
GB	6.2500	158.7500	8.5000	1.1250	.120	9	13,243
GC	6.5000	165.1000	8.7500	1.1250	.120	9	13,452
GD	6.7500	171.4500	9.0000	1.1250	.120	9	14,307

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TABLE I. BOUNDARY DIMENSIONS 1/ - CONTINUED

Bore Code	0B Bore Diameter (inch)	0B Bore Diameter (MM)	0D Outside Diameter	W Width	Chamfer Min. 2/	Dia. Series 3/	Dynamic Load Rating (lbs)
HA	7.0000	177.8000	9.5000	1.2500	.120	9	16,017
HB	7.2500	184.1500	9.7500	1.2500	.120	9	16,340
HC	7.5000	190.5000	10.0000	1.2500	.120	9	16,340
HD	7.7500	196.8500	10.5000	1.3750	.120	9	18,791
JA	8.0000	203.2000	10.7500	1.3750	.120	9	19,542
JB	8.2500	209.5500	11.0000	1.3750	.120	9	19,219
JC	8.5000	215.9000	11.5000	1.5000	.120	9	22,211
JD	8.7500	222.2500	11.7500	1.5000	.120	9	23,342
KA	9.0000	228.6000	11.9990	1.5000	.120	9	22,638
KB	9.5000	241.3000	12.7490	1.6250	.160	9	26,059
LA	10.0000	254.0000	13.2500	1.6250	.160	8	27,464

1/ All dimensions are in inches unless otherwise specified.
2/ The chamfer on bearings will clear a maximum fillet radius equal to the minimum chamfer.
3/ Diameter series is a soft metric conversion of the inch outside diameter and width in comparison with the boundary dimension in ANSI/AFBMA STD 20.

TABLE II. CAGES

CODE	CAGE
1	Pressed Steel or Molded Plastic
2	Machined Brass or Bronze
3	Machined Non Metallic (phenolic)
4	Pressed Brass

Table III. RADIAL INTERNAL CLEARANCE AND TOLERANCE CLASS

Code	Symbol	Tolerance
A	2 (Tight)	ABEC 1
B	0 (Normal)	
C	3 (Loose)	
D	4 (Extra Loose)	
E	2 (Tight)	ABEC 5
F	0 (Normal)	
G	3 (Loose)	
H	4 (Extra Loose)	

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TABLE IV. LUBRICANTS AND CLOSURES

Code	Lubricant	Closure
A B C	DOD-G-24508 MIL-G-81322 Preservative 1/	OPEN (FIG 1)
D E F	DOD-G-24508 MIL-G-81322 Preservative 1/	SINGLE SHIELD (FIG 2)
G H	DOD-G-24508 MIL-G-81322	DOUBLE SHIELD (FIG 3)
J K L	DOD-G-24508 MIL-G-81322 Preservative 1/	SINGLE SEAL (FIG 4)
M N	DOD-G-24508 MIL-G-81322	DOUBLE SEAL (FIG 5)
P R	DOD-G-24508 MIL-G-81322	SINGLE SHIELD AND SEAL (FIG 6)

1/ Preservative shall be in accordance with MIL-P-197.

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI Navy - OS

User Activities:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA - FSS

Preparing Activity:
DLA - IS

(Project 3110-0887-06)

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